

This manual is intended for reference purposes only.
Illustrations and technical specifications are subject to change without notice.



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GPS-Denied

Drone Navigation with Visual-Inertial Odometry



Precise Positioning

Accuracy Error $\leq 2\%$



Vision + IMU Fusion Navigation



No Map Required | Quick Deployment



N200M VIO Navigation Module



VIO Navigation Module

No GPS | No Map Required



VIO Loitering Munition System

Target Recognition & Autonomous Strike



VIO Swarm System

Multi-Drone Coordination



UNIBIRD is an Asia-Pacific innovator in drone technology R&D and manufacturing. We deliver high-performance, user-friendly AI modules that power the global drone industry.

By fusing embedded systems expertise with advanced edge computing and neural networks, our integrated solutions enable critical applications including VIO navigation, autonomous target recognition and tracking. UNIBIRD makes AI adoption accessible, accelerating large-scale deployment of intelligent modules to advance drone capabilities worldwide.

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N200M VIO Navigation Module

—Overcoming GPS-Denied Flight Challenges for UAVs

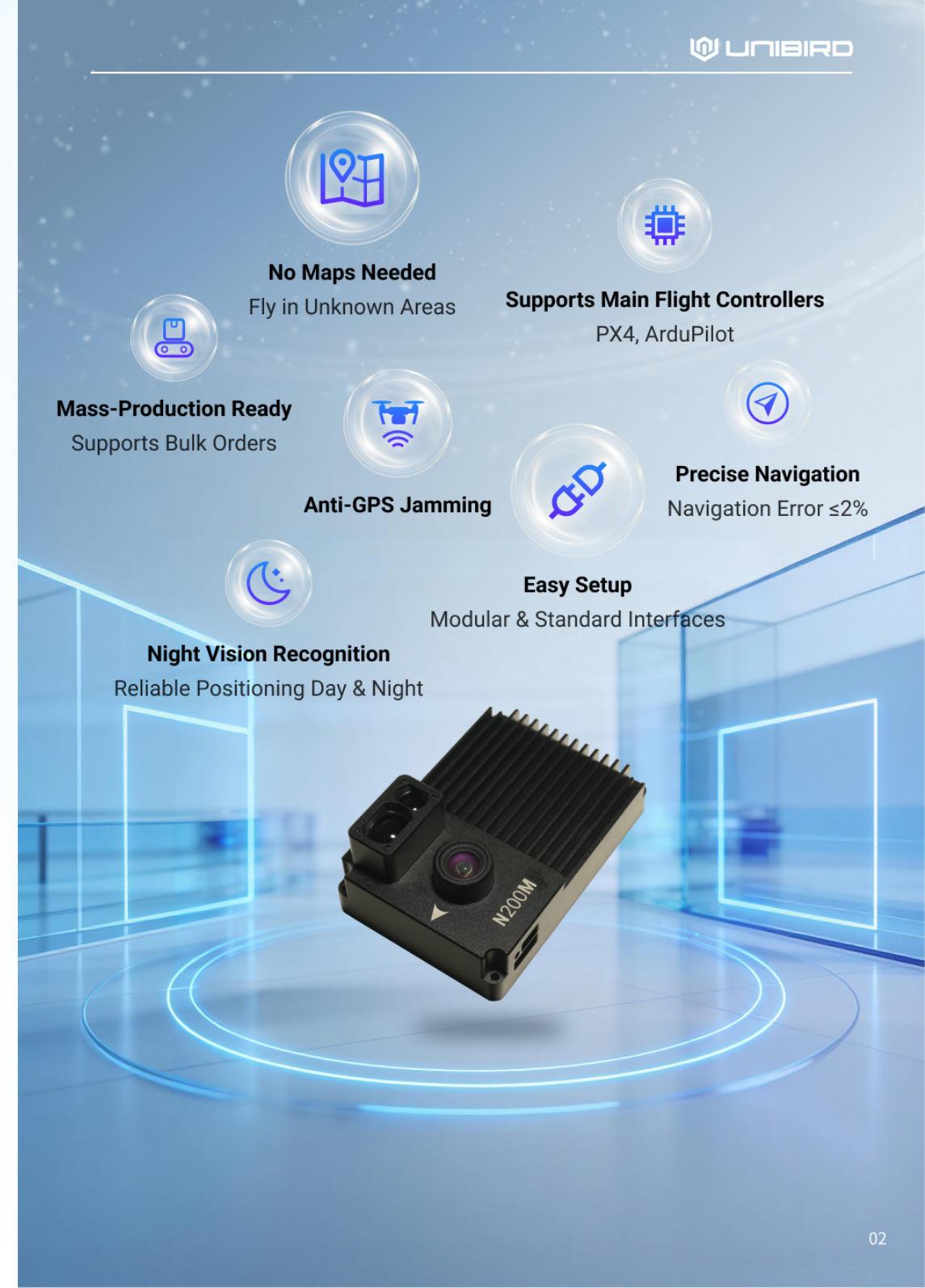
What is VIO?

VIO (Visual-Inertial Odometry) is a positioning technology that fuses camera imagery with an Inertial Measurement Unit (IMU). It calculates the drone's movement in real time by combining ground images with IMU data, enabling precise positioning. Even in environments with weak or jammed GPS signals, VIO ensures stable flight.

N200M VIO Navigation Module

The **N200M** is a drone positioning device based on VIO technology, designed to maintain normal flight even when GPS signals fail or are spoofed.

No preloaded satellite maps are required. The module can operate directly in any unfamiliar area with ground textures, enabling the drone to take off, hover, and follow flight routes without GPS.



Environmental Sensing



HD Camera Visual Sensing Unit

Capture Ground Features



Laser Rangefinder Auxiliary Sensing Unit

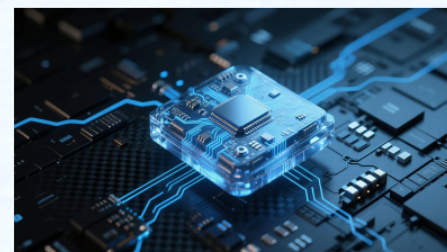
Accurate Distance & Height



Sense · Fuse · Position

VIO Visual Navigation Without GPS

AI Fusion Computation



AI Computing Board Core Computing Unit

Real-time Attitude & Accurate Positioning



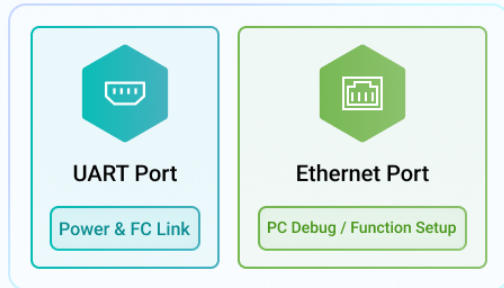
Drone Relative Position Output



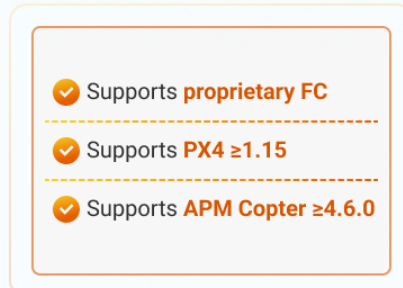
Quick Setup & Easy Use

N200M VIO Navigation Module –Efficient interfaces for easy drone integration.
No coding is required.

Interface



Supported Flight Controllers

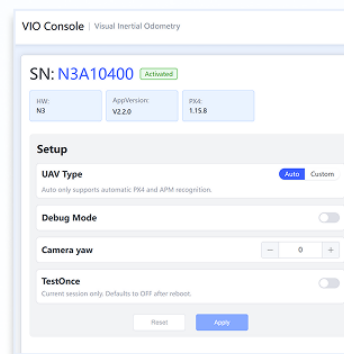


Web Console Setup

- Supports Web Interface, No Coding Required
- Simple Operation for Basic Testing & Parameter Setup



Cable Connection

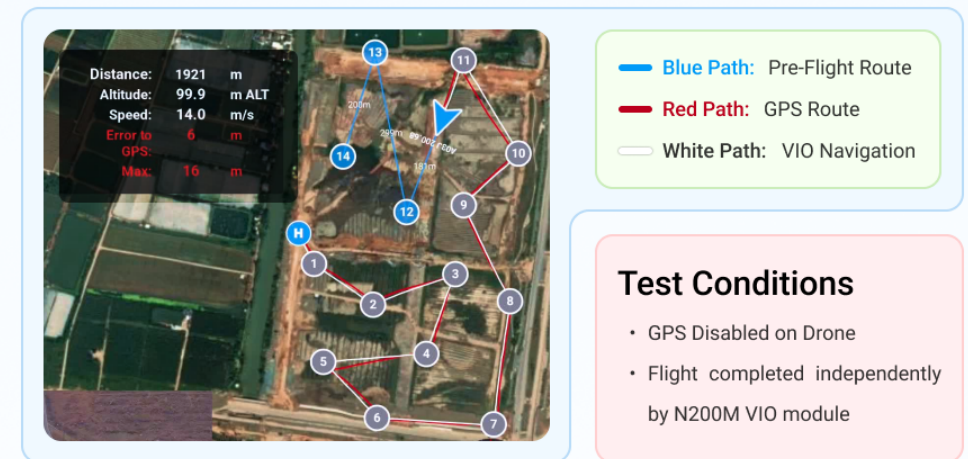


Web Setup Page

Tested Performance:

VIO Visual Navigation vs GPS Navigation

—Highly accurate (error ≤ 2%) positioning, matching GPS performance.



Test Log (Altitude: 100m)							
Distance (m)	196	412	993	1711	2943	3800	4750
Error to GPS (m)	3	5	9	15	40	61	75

N200M VIO Module Maintains
Stable, Accurate, and Reliable Positioning
Without GPS

*Data Based on Agricultural and Forestry Environment Test Results; Different Flight Conditions May Vary.

VIO Loitering Munition System

—The World's First Mass-Producible

Automated Reconnaissance and Strike Solution Designed
for GPS-Denied Environments

VIO Loitering Munition System consists of an integrated search-strike drone and a ground station.

The integrated search-strike drone is equipped with the **N200M VIO navigation module**, enabling stable operation under GPS-denied conditions or in electromagnetic interference environments.

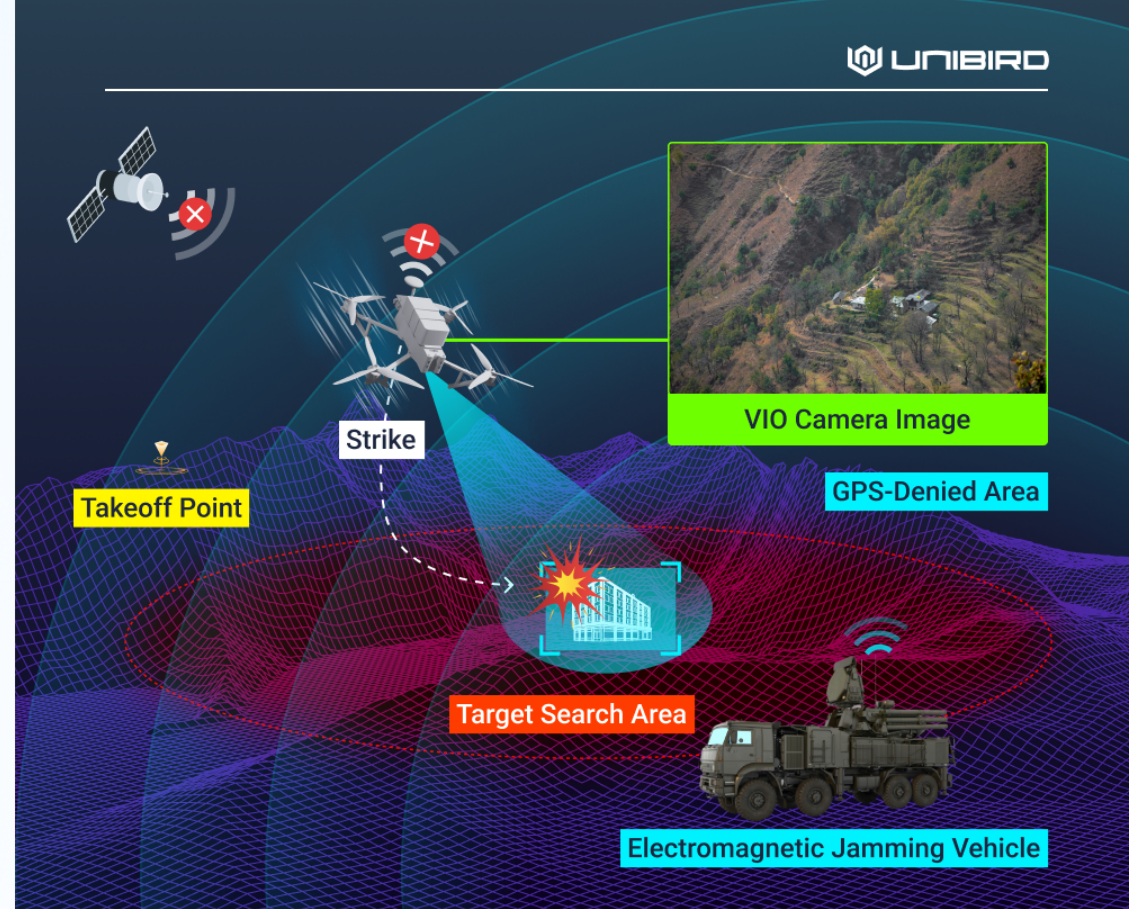
It supports an automated workflow from area patrol and target detection to mission execution :

- **VIO Navigation:** Visual-inertial odometry for precision flight in denied environments
- **AI Target Search:** Real-time recognition and locking during reconnaissance
- **High-Speed Strike:** Rapid response capability enhancing operational efficiency

Ground station specifications:



size	91*141*92mm
weight	1.3kg



◆ Key Features

Integrated Mission Workflow Integrated search-to-strike workflow.

Sensor-to-Shooter Efficiency

Utilizing onboard edge computing capabilities, It reduces "detection-to-strike" reaction time to seconds-level, significantly improving strike effectiveness against time-sensitive targets.

Automated Target Search

The system supports generating efficiently covered search paths via pre-set "route + rectangular area". Supports recognition of hundreds of target objects.

Supports self-training models

Users can train proprietary AI recognition models using their own datasets and import them.

Long-Range Precision Strike

Maximum Recognition Distance: 300 m
Maximum strike speed : 30m/s
Target Strike Accuracy: >98%

Ultra-long control distance

Maximum control distance reaches 15 kilometers.

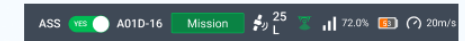
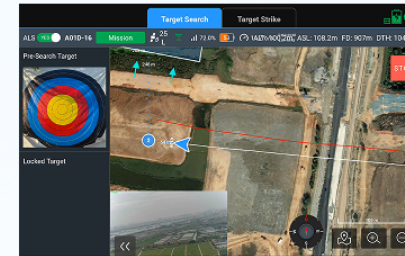
User-Friendly Operation

The world's easiest-to-operate drone, with one-touch takeoff and simplified controls. Efficient operation without requiring professional pilots.

◆ VIO Loitering Munition System Software

VIO Navigation

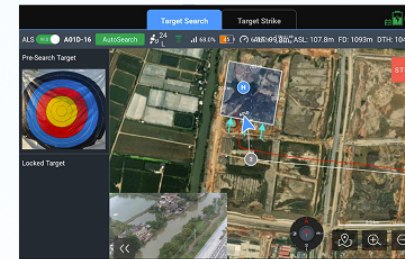
Seamlessly switches to VIO mode in GPS-jammed environments.



Near jamming area, GPS invalid ;
switching to VIO navigation in flight.

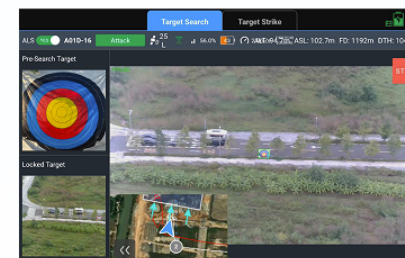
Target Search

Autonomously plans paths and traverses subdivided search areas.



Target Strike

Dynamically adjusts flight trajectory via the visual image for high-speed precision strikes without GPS.



VIO Swarm System

—The world's first combat-deployable swarm system operational in GPS-denied environments.

The Swarm drone system consists of an AI search drone, up to 15 AI strike drones, and a ground station.

Equipped with the **N200M VIO navigation module**, the drones enable intelligent search and strike operations in GPS-denied or GPS-disrupted environments, with autonomous target detection, real-time transmission, accurate tracking, and rapid target locking.



VIO Swarm System Composition



(1) AI Search Drone The AI Search Drone can automatically fly to a designated area for search and automatically identify potential targets.

(2) AI Strike Drone After receiving the target location and image information transmitted by the search Drone, the AI Strike Drone can automatically fly to the target location, identify and lock onto the target, and execute the strike mission.

Note: The AI Search Drone and AI Strike Drone use the same model of equipment. However, during mission execution, a separate Drone must be assigned to perform the AI search task, and it can return after the search is completed.



(3) Ground Station The Ground Station is the communication and control center of the VIO Swarm System, as well as a scheduling platform for integrating search and strike.

Key Features

Recon-Strike Coordination

Search drones autonomously search targets and relay data to ground stations; strike drones execute engagements.

Area Reconnaissance

Users define reconnaissance zones; drones autonomously plot routes and identify targets.

Smart Task Automation

The system automatically identifies targets, assigns tasks, and rapidly responds and executes.

Single-Operator Control

One operator controls up to 16 drones simultaneously.

Supports self-training models

Users can train proprietary AI recognition models using their own datasets and import them.

Ultra-long control distance

Maximum control distance reaches 15 kilometers, extensively expanding mission coverage.

Swarm Strike Modes

Supports multi-target distributed strikes or single-target concentrated strikes.

Product Specifications

N200M VIO Navigation Module Specifications



Standard



Night Vision

Category	Parameter	Details
Basic performance	Flight Altitude	30~200m
	Max Flight Speed	15m/s
	Navigation Accuracy Error	≤2%
	Output Frequency	30Hz
	Supported Flight Controllers	PX4、ArduPilot
	Operating Environment	Supports night vision, suitable for most scenarios
	Output Protocol	MAVLink

Drone Specifications



Category	Parameter	Details
Performance	Maximum Payload	3kg
	Maximum Strike Speed	30m/s
	Endurance (No Payload)	30min
	Maximum Recognition Distance	300m
	Maximum ControlRange	15km(Dependent Environment)
	Recognizable Target Types	Hundreds of categories including personnel, vehicles, cloth targets, etc. (recognition model training platform provided)
Visual Navigation Module	Navigation Accuracy Error	≤2%
	Max Flight Altitude	200m
Battery (Reference)	Recommended Battery	6S1P Li, 25c
	Capacity	10000mAh
Digital Video Transmission	Power Level	1.4G 25dBm±2
	Transmission Distance	15km(Dependnt Environment)
	RF Frequency Band	1427.9-1447.9MHz